MeatCODE - Adaptive Oracle memory loop

26 October 2026 · Proposed design for review; not deployed.

Purpose: Make Oracle progressively more useful to each user by remembering relevant goals, constraints and feedback, while keeping personal context separate from validated scientific evidence.

1. Capture the interaction

• Oracle sends the question, current context and explicit feedback to the Render API.

• The API records selected events in Neon SQL with user, project, time and source-message identifiers. Current-turn context can be used immediately.

2. Digest useful information

• A background process extracts durable preferences, project goals, constraints and decisions; it avoids retaining every sentence as permanent memory.

• Reconcile updates with existing memory. Distinguish explicit user statements from inferred preferences; uncertain or conflicting updates can request confirmation. This process is a proposed service, not a fourth fixed LLM.

3. Store private, structured memory

• A separate logical vault in Neon Postgres holds versioned memory records and links back to their source interactions. SQL is authoritative; optional pgvector embeddings support semantic lookup.

• Scope access to the correct user/project. Provide review, correction, deletion and retention controls, including removal from derived indexes. User statements remain user context, not verified scientific facts.

4. Adapt retrieval and reasoning

• At each question, the API selects a small relevant memory set. Current explicit instructions take precedence over older preferences.

• User context guides filters and ranking across keyword, vector and bounded graph retrieval. M3 receives both context and cited scientific evidence, clearly distinguished. Preferences must not suppress relevant contradictory findings.

5. Close the feedback loop

• Oracle returns an answer or proposed next action. The next user response supplies new constraints, corrections or usefulness feedback, which repeat the cycle.

• Evaluate whether evidence selection and recommendations improve using explicit feedback, citation quality and task outcomes. Do not assume repeated interaction finds an optimal route; action execution requires its own authorization.

Relationship to the existing architecture

• M1/Qwen extracts; M2 combines rules, model assessment and human review. Approved scientific claims live in Neon as active or reserve evidence. Unresolved claims remain in review.

• The artifact vault, reserve evidence and user-memory vault serve different purposes. Reserve retrieval policy remains open. Personalization changes retrieved context and ranking; it does not automatically retrain M1, M2 or M3.

Review choices: What should be remembered? How long? Which updates require confirmation? The new purple loop on the editable diagram shows the proposed flow; existing scientific ingestion remains separate.